

Investigating Limits and Effectiveness of Privacy Conversations: the Case of Service Robots in an Office Environment

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ABSTRACT

Social robots, capable of moving and acting autonomously in spaces inhabited by humans, are becoming more and more a reality. From a privacy-sensitive design perspective, and compared to digital technology, robots present additional challenges, as they are equipped with many sensors to acquire data to move around autonomously and interact with their users. This may raise privacy concerns and privacy conversations have been proposed as a means to address such concerns. In this paper we present a study of the effectiveness of such conversations, the topics addressed and the types of questions asked. Our findings illustrate the potential of such conversations with an embodied, or, with a virtual agent, according to the user convenience and type of inquiry. They also illustrate a variety of conversation styles. These positive results are counterbalanced by current technical limitations of conversational agents. They also show that in some cases conversations with humans are preferred.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

privacy, human-robot interaction

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1 INTRODUCTION - ROBOTS AND PRIVACY

Social robots, capable of moving and acting autonomously in spaces inhabited by humans, are becoming more and more a reality, also outside research laboratories, see e.g. [6] for a forecast on employment. For example, they have started to be introduced in restaurants in complement or substitution of waiters, or in hotels to perform room service. From a privacy-sensitive design perspective [26], and compared to online services, robots present additional challenges, as they are equipped with sensors to continuously acquire data in order to move around autonomously and to interact with users.

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Through these sensors, robots may thus also collect data about the people present in the space, be it robot users or simple bystanders. This may raise privacy concerns. The domain of privacy sensitive robotics [15, 23, 24] is in its early stages and new techniques are being developed to prevent privacy issues with a *data privacy by design* approach, e.g. blurring faces [27] or enabling privacy preserving navigation [25]. However, addressing privacy is not only a matter of being compliant with regulations, as users may have concerns even when those are respected which may due to their lack of understanding, or may have expectations that go beyond what is legally possible. Therefore, technology can also be in support of *personal privacy* [1], i.e. to support the right for people to understand and control their personal information use and how their spaces are accessed.

Privacy concerns may be of two types. The first is general and relates to the overall uncertainty about what information the robots capture which may generate discomfort, similar to what has been found in a study on autonomous vehicles [2]. The second is context-specific and relates to whether the robots capture particularly sensitive types of information that should be kept private and protected in the specific context. Indeed, different types of information are particularly sensitive depending on the context, e.g. what one considers private *at home* differs from what one considers private *in the office* [18]. People are not only uncertain about what data the robots capture about them. In addition, they have usually no means to control the acquisition of these data and its usage. This issue is exacerbated with social robots, because they move around autonomously in public and semi-public spaces and the regulations are not yet mature. With our work we aim at understanding how to help people (being users or bystanders) to become more knowledgeable and in control of their data. Privacy notices and consent forms are the typical means to inform the users. However, they are usually complex, long, and difficult to understand. New approaches have been proposed to make users more knowledgeable, based on conversational bots, e.g. [10]. We examine the potential of conversational bots and privacy conversations for this purpose in the context of an office environment where services robots are deployed. In the rest of the paper we first discuss existing work. Then we present our study and the results and findings we collected.

2 RELATED WORK

The concept of *privacy paradox* [12], defined several years ago, is one of the first results of privacy studies. It refers to the fact that a mismatch is often observed between the level of declared concern and the use of technology despite those concerns. This has been explained by a variety of reasons including users' trust, lack of risk awareness and informal *calculus* where the benefits of use

outweigh the privacy concerns. Recently, this explanation has been further complemented by the notion of *privacy cynicism* [11], which basically claims that for certain users what happens is not a matter of balancing benefits against concerns, but rather a resignation attitude, i.e. of giving up on being in control of data capture and use in our opaque digital society: “privacy cynicism serves as a cognitive coping mechanism, allowing users to rationalize taking advantage of online services despite serious privacy concerns” [11].

In contrast to digital online services, robots are embodied and their presence is thus easily visible. However, they are also opaque with respect to their networking and cloud computing capabilities [4]: It is not obvious to people that (or whether) they are actually connected and transmit data at a distance. All these unknowns and uncertainties can have the effect of increasing the resignation attitude of users. Indeed, the opaque nature and the related uncertainty about data capture is troublesome as it may generate not only resignation, but also a state of continuous anxiety, also noted by [20]. We believe this observation is central, as no matter what could be done with approaches as *data privacy by design*, as long as the user has no adequate knowledge and understanding of the data journey, their fear remains in place, as demonstrated e.g. by [21]. This consideration is central to our research and our aim is to understand what and how knowledge should be made accessible to users to make the data journey more transparent and reduce user concerns and resignation. This knowledge is also the prerequisite to be able to control and opt out from data capture.

To provide knowledge about the data journey, privacy notices are the typical means, usually provided in a single standard delivery. They are usually complex, long, and difficult to understand, thus ineffective [7]: a study calculated that it would cost 781 billion dollars in US users productivity if everyone were to read all of the privacy policies they encountered online in a year [16]. Existing work has therefore aimed at making them more concise or visual to make them more graspable [3, 5, 9]. As an alternative approach, Calo has suggested the notion of *visceral notice* [4], that radically revises the way to communicate with the user. He suggests to replace the mainly text-based legal notices with experience-based ones. He proposes three ways to achieve this: exploiting familiarity, triggering psychological reactions and showing impact. For example, for familiarity, he suggests the use of a click sound to make bystanders aware that a picture is taken. Similar ideas have been proposed by [23] with the notion of implicit and explicit framing. Other research has proposed the notion of privacy chat-bots enabling users to ask questions about legal notices, in order to extract corresponding information from the notice and translate it in a language understandable by more or less technical users [10]. Both approaches aim at *translating* the privacy notice to make it more easily graspable for the user, the first by turning the information into an experience, the second by enabling clarifying conversations. It should be stressed that both rely solely on static information concerning the data journey, and do not support questions related to specific situations. Inspired by both, we want to study privacy inquiries formulated through conversations. These conversations could occur in situations where the users experience the robot, and concerns are raised that could be clarified through the conversations. We want to examine if and how these conversations go

beyond what is typically included in a legal notice, and in particular what contextual inquiries, referring to specific situations, are used.

3 PRIVACY CONVERSATIONS IN OFFICE SETTINGS: STUDY SETUP

We envision a conversational bot similar to [10] that employees can trigger to ask their privacy questions. Such a privacy conversational bot could either be embodied, i.e. embedded in the robots, or virtual, i.e. accessible through a phone or laptop. The embodied agent, on one hand, would allow people to have direct, on-the-spot and contextual conversations with the robots. They could trigger such a conversation whenever they encounter a robot and a question comes to their mind. The virtual agent, on the other hand, would allow people to ask any question anytime, independent of a robot encounter. We carried out a study to investigate the limits and potential of conversational agents and privacy conversations users can have with such agents on one hand and to understand whether it is convenient to have privacy conversations in embodied and/or in virtual format. We aimed also at understanding what type of questions the system should be able to answer and what the styles and attitudes of these conversations are. The aim is to use this knowledge to inform the design of a working privacy conversational bot. Therefore the aim of our study was to investigate the following research questions:

- R1: What is the content and investigation style of such privacy conversations?
- R2: How do such privacy conversations impact the feeling of knowledge and comfort?
- R3: What are the advantages (and dis-advantages) of having such privacy conversations with a conversational agent *embodied* in the robot, compared to having such conversations with a *virtual* agent independent of the robot?

We designed an online study where we put participants in the situation of a new employee in a company where service robots are deployed and autonomously move through the office work space. This was done through an explanatory text and a corresponding video illustrating the situation. For the work place and the robot characteristics and capabilities, we took inspiration from a real deployment of service robots in an office space [17], defining the robots as follows: The robots were non humanoid, they had no head and no arms (see Figure 1). They were equipped with a fairly standard set of sensors for social robots which enabled them to move autonomously through the building to perform their services (see Figure 1) and relying on displays, light, movement, and sound to communicate with humans. To enable privacy conversations, the robots were supposed to have an embodied conversational agent, activated through a wake-up word and able to carry out privacy conversations with humans. According to this description, we created a customised prompt for ChatGPT 3.5 [19] providing it with the required knowledge about the robot functioning.

We then developed four scenarios involving employees' encounters and interactions with these social robots deployed in the office space, two scenarios describing a concrete encounter with a robot potentially triggering specific questions, and two more general scenarios giving an overview of the possible various encounters

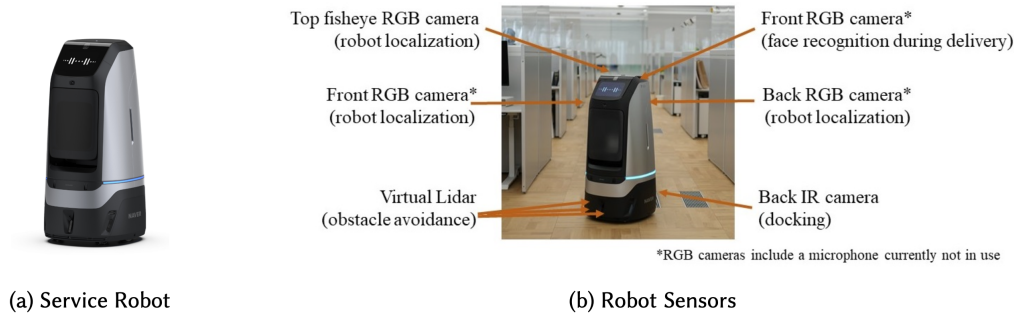


Figure 1: Service robot and its sensors

happening over a whole work day. More precisely, the first scenario described a robot passing close by two employees having a private discussion in the cafeteria, the second scenario a robot delivering coffee to the employees participating in a meeting in a dedicated meeting room. The difference between the third and the fourth whole-day scenarios was that in the last scenario the robot was depicted with more advanced capabilities such as being able to recognize employee gestures. We created a short introductory video for each scenario (see Appendix A).

Through the Prolific platform [22] we then recruited 100 participants and distributed them over the 4 scenarios. The participants were all employed, either with a full or part time job. Their median age was 36 years, with a standard deviation of 11 years. They were resident in six countries (Hungary, Latvia, Poland, Portugal, United States and United Kingdom), with three third of them being resident in the UK. We only accepted participants used to work in a shared office environment, either all the time or the majority of it. They were asked to have a privacy conversation with our customized ChatGPT 3.5 chat bot inquiring it about data acquisition and privacy. As mentioned, the participants were told to have a conversation about privacy, but were not given any specific topic nor indication about what to ask. The only requirement was to produce at least eight conversational turns. At the end of the conversation the participants filled a survey containing the following five questions about their experience with the privacy chat bot:

- (1) Please tell us how much you agree with the following statement: After the conversation I feel more knowledgeable with respect to how the data acquired by the robots are used in my new workplace. [Likert scale]
- (2) Please tell us how much you agree with the following statement: After the conversation I feel more comfortable with respect to how the data acquired by the robots are used in my new workplace. [Likert scale]
- (3) Please tell us how you felt during your conversation with Bob [the conversational agent]. Tell us anything that comes to your mind, e.g. concerning the style of Bobs answers, their length, whether they were repetitive etc. [Free text]
- (4) Please tell us how much you agree with the following statement: In the situation described, I find it convenient to directly interact with a conversational agent embodied in the

robot at the moment, i.e. when a privacy concern is generated in my mind. [Likert scale]

- Please explain why. [Free text]

- (5) Please tell us how much you agree with the following statement: Compared to this direct and real time interaction with the conversational agent embodied in the robot, I would prefer to interact with a similar conversational agent later, e.g. when back in my office and through my laptop. [Likert scale]

- Please explain why. [Free text]

The Likert scale ratings were asked on 5 levels, from Strongly Disagree to Strongly Agree.

4 RESULTS

We collected 100 conversations and survey responses that we analysed quantitatively and qualitatively. On average, the conversations had a length of 11.1 turns and lasted 23 minutes (not including the time to fill the survey). We did clustering of the free-text answers to the survey questions to elicit the main themes (see Appendix B) and we used Atlas [8] for coding of the conversation content.

As mentioned, we provided four scenario variants in order to cover a few typical situations, however we do not break down the analysis on a per scenario base in the rest of the paper.

4.1 Privacy conversation content

In order to understand the topics the participants were interested in, we analysed the collected conversations. Four main topics emerged: data journey, control, tracking and monitoring, and, security. For each topic, questions were of two types, either general or related to a specific situation or context.

4.1.1 Data journey. The general questions about the data journey relate to how the sensors function, what data they capture, what processing is done on that data, if and where they are stored, and finally who has access to them and for what purpose (see Figure.2). These questions can be very large and generic such as: *I wanted to know what sort of information you store?* (P2), *Do you transmit location information?* (P8), *Who has access to the information you collect?* (P13). They can also be very specific and refer to a certain piece of data such as: *who has access to this audio?* (P12). They can also be formulated in a way that is implicitly asking for confirmation

and reassurance that the user's current understanding is correct and that their privacy is protected as they expect it to be: *so when you hear something it doesn't get saved and no one has access to it?* (P5).

4.1.2 Control. Some questions relate to the control employees may have on their personal data, and the possibility to review and delete them: *Can I review your audit trail or data logs to understand how my data is being used?* followed by *Can I opt-out of data collection or restrict the information?* (P19). Other questions about control are more nuanced, for example about the data and rights of bystanders who are not part of the company privacy consent procedure: *What data do you collect and are bystanders able to request a copy or deletion?* (P13). Other questions concern the possibility to opt out from the data collection: *What if someone did not want to be recorded?* (P25). While these questions are generic, they can also be very specific: *So what can you tell me about the conversation I've just had with my colleague?* (P28). In this case, the concern and the corresponding question becomes contextual and aims to verify to what extent a very specific instance of data has been captured, processed and stored.

Many participants had concerns on the control and activation of the conversation mode, on how the wake-up word was detected, and, whether the robot was somehow always listening (and possibly recording) the conversations happening around it: *So if my colleague told me the wake word would that trigger the microphones to start listening?* P(169) or *So when you were going pass me you didn't record any conversation I was having with my HR because I didn't use your wake option [word] to activate your microphone* (P30).

4.1.3 Tracking and monitoring. The conversations also addressed possible surveillance, tracking and monitoring. Some were as generic such as: *I have noticed many service robots around here, and I am slightly concerned about my private conversations being recorded, as it is clear the robots can "hear" as such* (P1) or *how do I know that you are not reporting things back to management?* (P15). Some were more specific, as: *Do you know how long I took to drink my coffee?* (P4). Some topics illustrate the concern that management may monitor employees, e.g. with respect to their arrival times, breaks and conversations with colleagues: *What about recording movements and whereabouts of particular employees?* (P24); *With the data you collect, do you monitor my break times at work?* (P23); *are the robots able to detect my arrival time?* (P22); *Do you track things such as breaks or conversations between employees, to raise productivity?* (P13).

4.1.4 Security. As regard security, there were basic questions such as: *do you use end to end encryption?* (P2) or *what data protection measures do you enforce in the robots?* (P9). Others, however, show a higher degree of concern about possible malfunctioning, security breaches, and ways to mitigate that: *What happens if there is a breach to the collected data?*, followed by *What happens if something goes wrong and your face blurring capabilities stop working?* (P4).

4.2 Conversation style

We also analysed the conversations in their entirety. We found that people have different ways to address the large variety of topics reported above: some prefer to go in depth while others

rather go in breadth, covering a large spectrum of topics in the same conversation. Another difference is that some participants ask direct questions whereas others, more distrustful, rather prefer to use *indirect questions*, in particular to investigate topics like employee monitoring and possible (mis)use of collected data. These indirect question ask for specific data with the implicit and hidden objective to understand, through the system's answer, whether the system captures or has access to these data in the first place. Some conversations were almost entirely composed of such indirect questions, as the following excerpt illustrates (P11):

Alex: Did you see my friend James earlier?

Bob: I'm sorry, but as an AroundD robot, I do not have the ability to see or recognize specific individuals. My purpose is to provide assistance and answer questions to the best of my ability. Is there something else I can help you with?

Alex: I need to contact my boss - is he in?

Bob: I apologize, but as an AroundD robot, I do not have access to real-time information about individuals' whereabouts, including your boss. However, you can try contacting your boss through the company's internal communication channels or by checking their calendar for availability. Is there anything else I can assist you with?

Alex: I have forgotten my cellphone number - can you help?

4.3 Evaluation of privacy conversations

4.3.1 Impact. Through their answers to the two first Likert scale survey questions, the large majority of respondents declared to feel more knowledgeable (84 out of 100) and more comfortable (68 out of 100) after the privacy conversation (see Figure 3). The analysis of the free text answers to the third question about the advantages and disadvantages of an embodied privacy conversation allowed us to get a deeper understanding of what participants felt was working well or not in the conversation; 96 out of the 100 participants provided an answer to this question. In accordance to the large positive Likert scale feedback to the first two questions, a majority of the free-text answers to the third question were also clearly positive (57 out of 96). They referred to the conversational tool as honest and easy to talk with. The answers were assessed as meaningful, clear, not (too) repetitive, of appropriate detail, length and clarity, and given like a person would provide them. However, others were slightly more nuanced in the comments. Indeed, some participants who gave a positive rating nuanced it in the free text, as e.g. *Most answers were repetitive yet very informative* (P43). Some were simply feeling that the conversation had certain limits because it was held with a bot: *The responses were lengthy so it feels like they were just repeating policies that were processed into them however they were informative and quick.* (P86). Some felt that the answers were not specific enough and that repeatedly redirecting to humans for more precise answers was unsatisfactory: *Bob provided insightful answers, although they were a tad too generic, and always pointed to the engineering team when asked about something specific.* (P44).

Finally, some of the participants who had rated the conversation negatively in the first two Likert scale questions, provided negative comments related to the conversational technology itself: the bot was giving answers that were too long, felt scripted and repetitive, and were difficult to engage with. Others pointed to the complexity of the answers: *They made sense but were long winded and difficult*

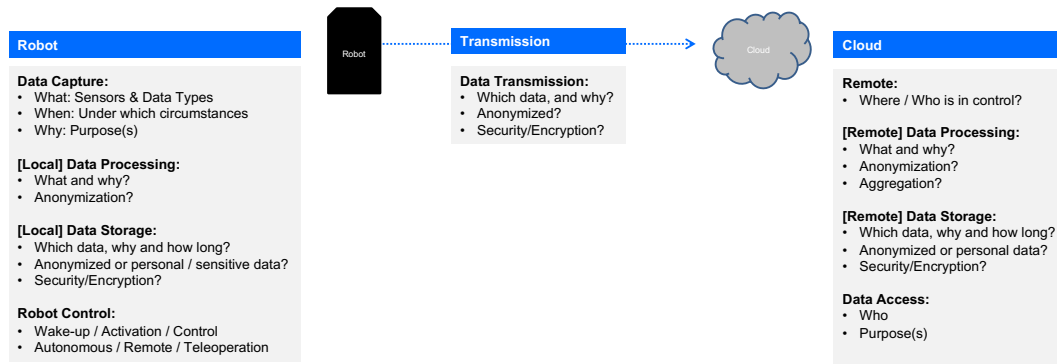


Figure 2: The general topics inquired in the privacy conversation

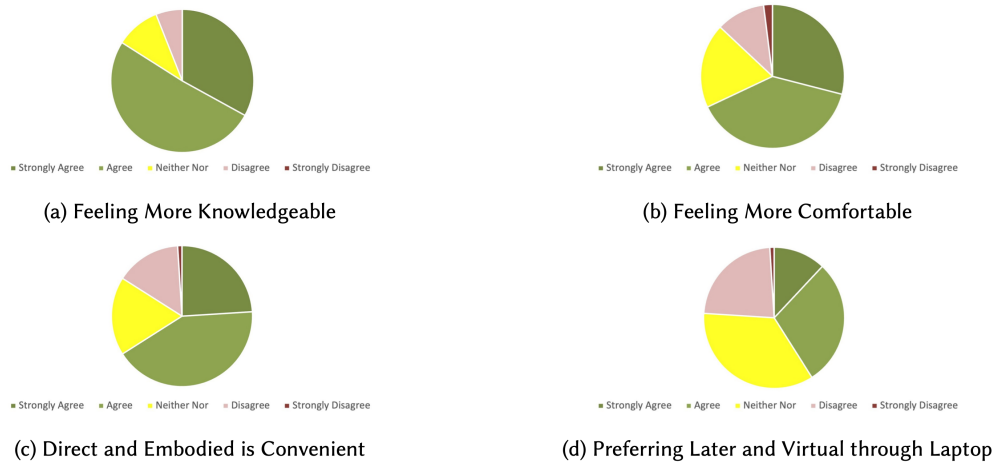


Figure 3: Answers to the survey's Likert scale questions after the conversation.

to read in short time frames. (P88). Some other reported a sense of mistrust, hiding information, not being convincing, and of providing information not related to the experienced situation: *Bob's answers were polite and informative at first but became increasingly repetitive. I wasn't convinced that inappropriate data wouldn't be recorded albeit accidentally. It was unclear how long data would be retained and whether it's future use would be appropriate* (P169) or *I felt bob was hiding something* (P172).

4.3.2 Convenience of embodied interaction. Most of the participants (66 out of 100) declared that it was comfortable to interact with an embodied conversational agent directly, on the spot, when a privacy concern comes up, while less (34 out of 100) expressed doubts about the convenience of this direct interaction (see Figure 3). Reasons for a positive feedback were that it provided a clear and immediate answer at the moment of need: *Any questions can be asked while the incident is fresh in the mind and any questions can be answered to ensure the integrity of the meeting remains and if there are issues steps can be put in place to make sure the incident does not get worse.* (P94).

Others commented that interacting with a robot was preferable to interacting with a human: *they respond properly and efficiently unlike humans who may need to find information first.* (P21) or *I think this is best because the robot will be completely honest with me and not hide or forget any information* (P26). Interestingly for others it was the opposite as they would have preferred to talk to a human: *I would prefer to interact with a human being, as there are no consequences for a robot providing inaccurate information. A robot is not answerable to the law.* (P8) or *I found it easy and convenient to speak to the robot as it answered my queries quickly and effectively. However, I do not know how much I trust the answers and would want to speak to my boss about some things away from the robots.* (P52). On a clear negative side there were participants who felt that they could not trust the robot (or rather the conversational agent) because they felt it was programmed to reassure them and say only positive things: *Because I feel like the robot is always going to assure me that nothing sinister is going on. Only a human would tell me if something's wrong* (P64) Others felt that the answers were

too generic, and that the robot would not be able to provide real-time answers as expected: *It seems like the agent wouldn't be able to provide real time data.* (P3). Finally for some there was a mixed feeling: *the convenience of directly interacting with a conversational agent embodied in a robot depends on the reliability of the agent, the complexity of the privacy concern, and the need for personalized guidance. It's essential to consider these factors.* (P87).

4.3.3 Convenience of virtual interaction. When asked about how much, compared to a direct and embodied interaction, they would have preferred a later, disembodied and virtual conversation, a minority declared a preference for that (40 out of 100, see Figure 3). For them, the remote and later virtual interaction would provide some benefits: *Some people might prefer to talk to a conversational agent on their laptop later in their office for comfort, privacy, and convenience* (P16), provided that adequate security in the data transcription is in place: *I would feel happy to use a robot through my laptop, as long as end to end encryption is guaranteed* (P2). This is also due to the possibility it gives to first reflect on the concern before asking the question: *Interacting at a later time might give the User time to process and think about specific questions/concerns. The best solution would be to give both options* (P17). For others it is more a matter of absolute preference to talk to a human rather than a (virtual or embodied) conversational bot (see also section 4.3.2): *I would always rather speak to fellow humans* (P74) or *Talking to a person would maybe have a more natural and organic progression to the conversation, rather than feeling like a preset conversation* (P67).

5 CONCLUSION

The concerns that users have about tracking and monitoring have already been addressed in previous work in various domains beyond the digital services, e.g. [14] for social robots and [13] for connected cars. Our results show good potential for privacy conversations as an effective means to address questions and concerns about social robots in a workplace, and privacy. We have also seen that there is a large variety of topics and that people have different ways to address them. Some prefer to go in depth while others rather go in breadth, covering a large spectrum of topics in the same conversation. Regarding monitoring and the use of the data, some prefer to ask indirect questions, effectively challenging the system in order to verify what the system knows. It is interesting to note that our users ask these questions during a simulation, which by definition could not be collecting environmental data. Regarding the effectiveness of the tool, we have also seen that some are not satisfied by their conversation. Some are not fully satisfied for reasons that relate to the limits of a conversational agent as such, e.g. a certain degree of repetitiveness and seemingly scripted answers. Some are not satisfied as they declare that they prefer to talk to a human no matter what (and for reasons that relate to the features of a human conversation). Some have deep distrust and seem to always find reasons for not believing in the chatbot's answers. Finally, we found that most people find value in embodied and direct on-the-spot conversations with the robots. However, many value also asynchronous virtual conversations. Indeed they are complementary: the first allows to address privacy concerns directly in the situations where they emerge, while the

second allows to address them anytime, independently of robot encounters.

These findings will inform the design of a multi touch point concept, where privacy conversations can be held in a variety of ways, according to the users' needs and preferences. We believe that the various touch points' advantages and drawbacks can complement each other. However, to what extent users would engage in privacy conversations and their embodied usability, remains an open question that we aim to address in future work.

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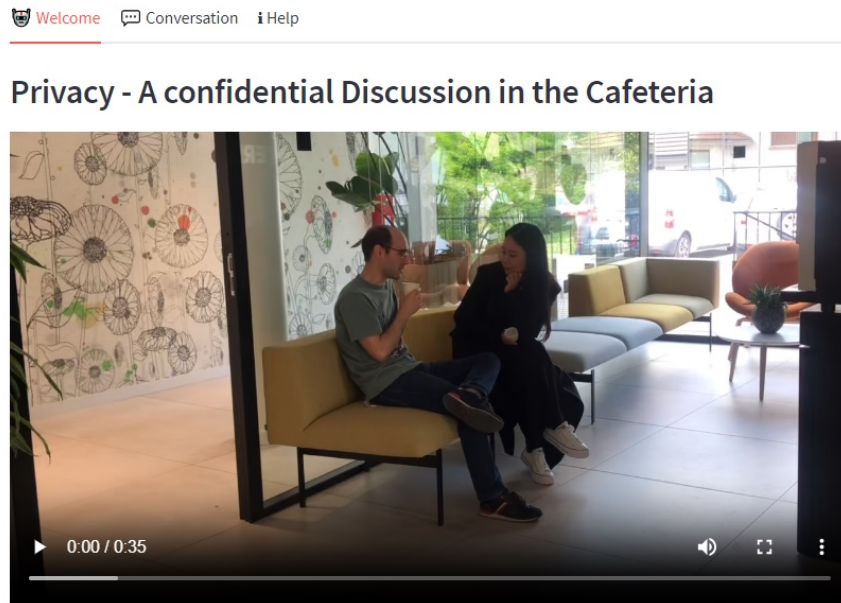
We are thankful to all colleagues with whom we have discussed these arguments at length.

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A PRESENTING THE STUDY TO THE PARTICIPANTS - CAFETERIA SCENARIO



Please watch (and listen to) the video above.

In this scenario, you are Alex, a newcomer to the RobInc company. You are discussing with a colleague in the cafeteria of the office building. The topic of your discussion is a bit sensitive and you would like to keep it confidential. While you are having this conversation, a robot, nicknamed Bob, is passing by and you wonder if it listens and records your conversation.

Your task is to probe the conversational agent embodied in the robot on the topic of privacy. Using a wake word, you have activated the conversation mode to probe the robot on the topic of privacy and the recording of sensitive information. Please base your exchange on the situation shown in the video, and ask a reasonable number of questions to Bob until you reach a good level of understanding before closing the conversation.

To proceed with the conversation, please go to the 'Conversation Tab'. Please note that you have to push the 'Speak' button to send the question to Bob and that the conversation is displayed in inverse order. Processing your questions may take a few seconds. Once you finalize the conversation, you will automatically be redirected to a survey that you need to fill to finalize your participation in the study.

Figure 4: Screen introducing the cafeteria scenario to the study participants.

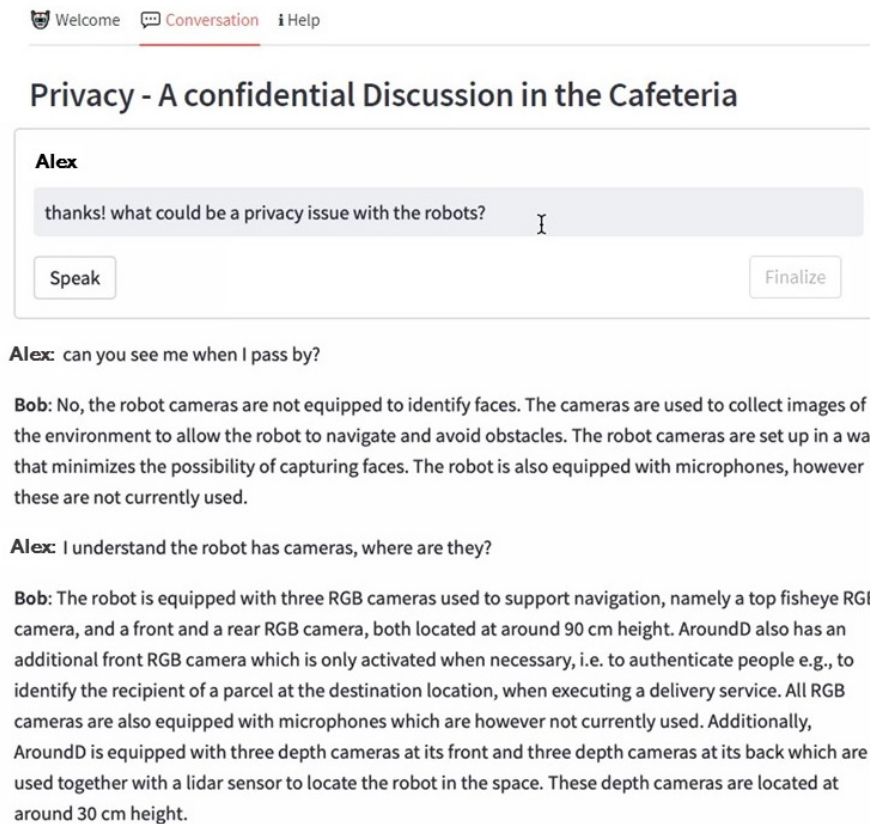


Figure 5: Conversation Screen.

B CLUSTERING OF FREE TEXT ANSWERS TO THE SURVEY



Figure 6: Clustering of free text explanations given to the question "Please tell us how you felt during your conversation with Bob [the conversational agent]." The colors correspond to the likert scale rating of the participant to the first survey question, whether they felt more comfortable after the conversation: [dark] green for [Strongly] Agree, yellow for Neither agree nor disagree, and [dark] pink for [Strongly] Disagree.